INDUSLINK: A System to bridge the Gap between Undergraduates and Industry

Project ID: 25-26J-281

Project Proposal Report

Almaas M.A.A – IT22172778

B.Sc. (Hons) Degree in Information Technology Department of Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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INDUSLINK: SKILL DEVELOPMENT, INTERVIEW PREPARATION AND PROGRESS TRACKING

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Almaas M.A.A IT22172778

Supervisor: Dr. Mr. Suresh Niroshan
Co-Supervisor: Ms. Fathima Fanoon

B.Sc. (Hons) Degree in Information Technology Department of Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Almaas M.A.A	IT22172778	

The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

The above candidate is carrying out research for the undergraduate dissertation under my supervision.

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Signature of the supervisor:

(Mr. Suresh Niroshan) Date: ………………

……………………………

Signature of the Co-supervisor:

(Ms. Fathima Fanoon) Date: ………………

Abstract

The transition from undergraduate education to the industry presents a significant challenge for many students, who often face skill gaps, limited interview preparation, and a lack of structured self-assessment mechanisms. Existing employability-enhancement systems have primarily focused on isolated interventions such as technical workshops or mock interviews without providing a holistic, adaptive, and continuous support framework [1], [2]. To address this gap, the proposed research introduces a student-centered digital framework that integrates system-guided skill development plans, mock interview generation tailored to job clusters, self-assessment and progress tracking dashboards, and a social feed for peer learning and experience sharing.

The research scope focuses on mapping job skill requirements to student profiles through the use of Natural Language Processing (NLP) and machine learning, enabling accurate identification of gaps between industry expectations and student capabilities. Based on these insights, adaptive learning paths are designed to dynamically adjust in response to individual student needs, ensuring that skill development is both targeted and personalized. In addition, the study emphasizes evaluating student progress and overall system effectiveness through reflective self-assessment tools, feedback analytics, and survey-based validation. By aligning student competencies with evolving labor market requirements, the proposed system aims to enhance not only technical preparedness but also structured interview readiness, while fostering sustained motivation through peer interactions.

The expected outcome is a scalable employability readiness framework that improves undergraduate students' industry alignment, enhances confidence during job interviews, and promotes self-directed lifelong learning. Beyond academic contribution, this framework supports universities and policymakers in designing evidence-based interventions to reduce the academic–industry readiness gap in IT education.

Keywords: Skill Development, Interview Preparation, Progress Tracking, Self-Assessment, Employability Readiness, Peer Learning, Adaptive Learning Paths, Industry Academia Gap.

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List of Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency-Inverse Document Frequency
BERT	Bidirectional Encoder Representations from Transformers
WBS	Work Breakdown Structure
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
LMS	Learning Management System
API	Application Programming Interface
GUI	Graphical User Interface
KPI	Key Performance Indicator
CSV	Comma-Separated Values

1. Introduction

1.1 Background & Literature Survey

The transition from higher education to the industry remains a major challenge for undergraduates worldwide. While many graduates possess adequate technical knowledge, studies consistently highlight gaps in employability skills, structured interview preparation, and mechanisms for self-assessment (Suleman, 2018; Finch et al., 2013). Employers increasingly emphasize the importance of communication, teamwork, adaptability, and problem-solving, yet universities often provide limited structured pathways for students to develop these competencies. This persistent industry–academia gap underscores the need for innovative frameworks that combine technical readiness with holistic professional development.

Several interventions have attempted to bridge this gap, including technical workshops, soft-skills training, and mock interviews. Research has shown that mock interviews and targeted question banks can improve students' confidence and performance (Robles, 2012). However, many of these approaches are static, offering one-time exposure rather than continuous preparation. Likewise, progress monitoring and reflective self-assessment critical tools for sustained growth—are often absent from existing career preparation programs (Yorke, 2006). Without structured feedback and adaptive pathways, students struggle to track improvements and adjust their strategies effectively.

In addition, recent literature highlights the growing role of peer-driven learning environments, where students benefit from sharing resources, experiences, and feedback with one another (Boud, Cohen, & Sampson, 2014). Yet, current employability frameworks rarely integrate such social or collaborative features, which could sustain student engagement and motivation. Taken together, prior research highlights the value of skill development, interview practice, and self-assessment but lacks an integrated, technology-driven solution. This study addresses the gap by proposing a holistic framework that combines system-guided skill development, adaptive interview preparation, progress tracking, and peer-driven social engagement to better prepare IT undergraduates for industry challenges.

1.2 Research Gap

Most existing employability initiatives for undergraduates focus on isolated activities such as short-term workshops or one-time mock interviews. While useful, they do not provide a continuous and structured pathway for students to develop the competencies needed for industry. Current platforms also rely on static interview materials and rarely include personalized skill development plans that adapt to individual needs. In addition, self-assessment and progress tracking tools are often missing, leaving students without measurable feedback to monitor their growth. Finally, although peer learning is proven to enhance motivation and engagement, it is largely absent from employability systems. These gaps highlight the need for a holistic, technology-driven framework that combines guided skill development, adaptive interview preparation, progress tracking, and peer collaboration in one integrated solutions.

Furthermore, the majority of available platforms fail to incorporate real-time feedback mechanisms**.** For example, students may complete exercises or mock interviews but rarely receive immediate, actionable insights into areas requiring improvement. This delay prevents timely correction of mistakes and slows skill acquisition. A system capable of providing instant feedback and dynamically updating learning paths based on student performance would better support continuous improvement and engagement.

Another practical gap lies in the alignment with evolving industry requirements**.** Job descriptions, required skills, and recruitment trends change rapidly, yet most employability tools rely on outdated or generic content. As a result, students are often preparing for roles that no longer match the current market needs, reducing their competitiveness. Integrating AI-driven analysis of job postings and industry trends can ensure that skill development and interview preparation remain relevant and targeted.

Research Gap Feature	Jackson et al. [1]	Andrews & Higson et al. [2]	Robles et al. [3]	Yorke [4]	Boud et al [5]	Proposed Research Solution
System-guided skill development plans	✗	**✓ (general soft skills)**	✗	✗	✗	✓
Adaptive interview preparation (role-specific mock Qs)	✗	✗	**✓(soft skills)**	✗	✗	✓
Continuous self-assessment & progress tracking	✗	✗	✗	**✓(conceptual)**	✗	✓
Peer learning / social engagement for employability	✗	✗	✗	✗	**✓ (peer learning)**	✓
Peer learning / social engagement for employability	✗	✗	✗	✗	✗	✓

Table 1.1

1.3 Research Problem

IT undergraduates often face significant challenges when transitioning from academic studies to professional employment. Despite possessing technical knowledge, many students struggle with employability skills, structured interview preparation, and self-awareness of their strengths and weaknesses. The lack of a systematic approach to developing these competencies leaves students underprepared for the demands of the industry.

Existing interventions, such as workshops, mock interviews, and short-term training programs, tend to be fragmented and static. They do not adapt to individual student needs, nor do they provide continuous feedback on skill improvement. As a result, students are unable to track their progress effectively or focus on areas that require targeted development, limiting the overall impact of these initiatives.

Additionally, current employability systems rarely incorporate peer learning or social collaboration, which are essential for motivation, engagement, and shared learning experiences. Without these interactive elements, students miss opportunities to learn from peers, exchange insights, and sustain preparation over time. Collectively, these limitations point to the need for a comprehensive, technology-driven framework that integrates personalized skill development, adaptive interview preparation, progress tracking, and peer-driven engagement to improve undergraduates' industry readiness.

Many undergraduates also express feelings of uncertainty and anxiety as they approach graduation, not because they lack technical competence, but because they are unsure how to present their abilities effectively in interviews or adapt their skills to industry expectations. This emotional gap, often overlooked in existing systems, contributes to reduced confidence and increased stress during the job search process. A framework that recognizes both the technical and psychological aspects of employability can better prepare students to enter the workforce with confidence.

Furthermore, the absence of continuous mentorship and peer support leaves many students feeling isolated in their career preparation journey. Real-world employability is not built in isolation but through collaborative learning, shared problem-solving, and exposure to diverse perspectives. By embedding social collaboration into employability training, students can motivate one another, learn from shared experiences, and sustain long-term engagement. This human-cantered approach ensures that employability solutions are not only technically effective but also emotionally supportive, preparing undergraduates for both the challenges and opportunities of the industry.

2. Objectives

2.1 Main Objective

In achieving this objective, the research intends to leverage advanced techniques such as Natural Language Processing (NLP), knowledge representation, and adaptive learning algorithms to ensure that the system is not only personalized but also responsive to dynamic labour market demands. This approach will allow the framework to align individual student profiles with evolving job requirements, ensuring greater relevance and practical applicability in real-world recruitment scenarios.

Furthermore, the objective extends beyond technical skill-building to emphasize holistic student development. By integrating reflective self-assessment, structured feedback, and social collaboration, the framework will foster both individual growth and collective learning. This ensures that undergraduates are not only technically proficient but also well-rounded professionals who possess the confidence, adaptability, and collaborative mindset necessary for long-term career success in the industry

2.2 Specific Objectives

In addressing these objectives, the study emphasizes the importance of personalization and adaptability in employability training. Rather than offering a static, one-size-fits-all solution, the framework will continuously adjust learning paths and interview preparation content according to evolving industry standards and the unique progress of each student. This ensures that students remain aligned with real-world demands and can focus on the most relevant areas of improvement.

Moreover, by embedding evaluation mechanisms such as survey-based validation and reflective analytics, the research aims to ensure that the framework is both evidence-based and student-centered. These feedback-driven refinements will not only improve the system’s usability and effectiveness but also provide a structured foundation for future employability-focused innovations. Ultimately, the specific objectives converge toward creating a holistic solution that empowers undergraduates to confidently transition into professional roles with both technical and soft skills well developed.

3. Methodology

Data Collection:

The first phase of the research focuses on collecting relevant data to understand both industry requirements and student readiness. Job market data will be gathered from local and global Ijob postings to identify essential technical and soft skills. Student data will include academic records, resumes, skill self-assessments, and internship experiences to evaluate existing competencies. Additionally, feedback from alumni and faculty will provide insights into real-world expectations, while peer collaboration data from group discussions or forums will help assess social learning dynamics. Collectively, this diverse dataset ensures that the framework is grounded in both academic and industry realities.

Data Preprocessing:

Once collected, the data will undergo preprocessing to make it suitable for analysis and modeling. Textual data from job descriptions, interview questions, and student profiles will be cleaned using techniques such as tokenization, stop-word removal, stemming, and lemmatization. Feature extraction will be performed using NLP methods like TF-IDF and BERT embedding to capture semantic relationships between skills, job requirements, and student competencies. Additionally, data normalization and formatting will standardize inputs, enabling accurate analysis of skill gaps and preparation needs.

System Design and Algorithm Development:

The core of the research involves designing a technology-driven framework composed of several interconnected modules. The skill development module will generate personalized learning plans using machine learning algorithms, adapting recommendations to individual student gaps. The interview preparation module will leverage NLP to produce dynamic mock interview questions tailored to targeted job roles. A progress tracking and self-assessment module will provide dashboards for monitoring daily, weekly, and semester-long improvement, helping students reflect on their learning journey. Furthermore, a peer learning module will support collaborative engagement through social feeds, where students can share experiences, insights, and achievements, enhancing motivation and community learning.

Evaluation and Validation:

The framework's effectiveness will be evaluated through a pilot study with final-year IT students. Quantitative measures, including skill assessments, mock interview performance, and progress-tracking metrics, will assess improvements in employability readiness. Qualitative feedback from surveys, interviews, and focus groups will evaluate usability, engagement, and perceived benefits, guiding iterative refinements to optimize the system's adaptability and student-centered features.

4. Project Requirements

4.1 Functional Requirements

The proposed framework will provide personalized skill development plans that generate adaptive learning paths based on individual student skill gaps and the requirements of the job market. This ensures that students receive targeted guidance on which skills to improve and in what order, creating a structured preparation journey.

To enhance employability readiness, the system will include mock interview preparation capabilities. Students will receive role-specific interview questions, practice their responses, and track their performance over time, allowing for continuous improvement and confidence building.

The framework will also support progress tracking and self-assessment, enabling students to monitor their skill development, visualize milestones, and receive feedback on their performance. These dashboards provide actionable insights and help learners adjust their strategies based on measurable outcomes.

To foster engagement and collaborative learning, the system will integrate peer learning and social collaboration features. Students can share experiences, discuss challenges, and post achievements through a social feed or discussion forum, motivating one another and facilitating knowledge exchange.

Additionally, the framework will provide analytics and reporting tools for both students and instructors, offering visual dashboards and reports to evaluate learning progress, engagement, and areas that need attention. Notifications and reminders will help maintain consistent participation by alerting students to upcoming tasks, suggested activities, or milestones. Finally, integration with job and student profiles ensures that skill development and interview preparation are aligned with industry requirements, making the framework comprehensive and student-centered.

4.2 Non-Functional Requirements

The proposed framework must be reliable and available, ensuring that students can access skill development plans, mock interviews, progress tracking, and peer collaboration tools at any time without system downtime. It should be scalable, capable of supporting multiple users simultaneously, including large cohorts of students and faculty, without compromising performance.

Usability is a critical requirement; the system should have an intuitive and user-friendly interface that allows students to navigate learning modules, dashboards, and social collaboration features with ease. Performance is also essential, as the system should process data, generate personalized recommendations, and update dashboards quickly to provide real-time feedback.

The framework must ensure security and privacy, protecting student data, assessment results, and personal information from unauthorized access. Maintainability and flexibility are important so that the system can be updated with new learning modules, job requirements, or interview question sets as industry demands evolve. Finally, the system should support interoperability, enabling integration with existing academic tools, learning management systems, and external job portals to provide a seamless user experience.

5. Budget and Budget Justification

Item	Estimated Cost (LKR)	Justification
Cloud/Hosting Services (3 months)	3,500	To deploy and test the prototype framework for skill development and interview preparation modules.
Data Collection & Survey Tools	2,000	To conduct surveys/interviews with students and collect feedback for evaluation of progress tracking features.
Software Tools (ML/NLP libraries, IDEs)	2,500	For developing adaptive learning paths, mock interview question generation, and text analysis. Most tools are free, but some require premium plugins/APIs.
Hardware	3,000	For additional storage, internet access, and minor hardware needs during development and testing.
Documentation, Printing & Binding	2,000	To prepare research proposal drafts, progress reports, and final submissions.
Miscellaneous/Contingency	1,500	For unexpected expenses (e.g., additional APIs, domain registration, or extra survey costs).

Table 5.1

6. Anticipated Conclusion

The proposed framework is expected to demonstrate that a structured, AI-driven approach can provide IT undergraduates with clear and personalized pathways for skill development and interview readiness. Unlike existing fragmented solutions, the system aims to offer continuous guidance that adapts to each student's unique gaps, ensuring they are better prepared to meet industry expectations.

It is also anticipated that progress tracking and self-assessment tools will enhance student confidence by making growth measurable and transparent. Through reflective evaluation, students will be able to identify strengths, address weaknesses, and adjust their preparation strategies more effectively, leading to improved employability outcomes.

Finally, the inclusion of peer collaboration features is expected to create a supportive and motivating environment where students can share experiences, learn from one another, and sustain long-term preparation. Collectively, the research will conclude that this integrated framework not only bridges the industry–academia gap but also equips undergraduates with the technical and soft skills necessary to transition smoothly into professional roles.

In addition, the research is expected to highlight the value of incorporating advanced technologies such as Natural Language Processing (NLP) and adaptive learning algorithms in employability-focused systems. By leveraging these methods, the framework will not only align student skills with dynamic job requirements but also contribute new insights to the design of AI-powered educational tools. This ensures that the project has both academic relevance and practical significance for the industry.

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Appendices

[If any, include here.]